Quiz 1 — Sampling and NumPy

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices · same skills as Practice 2

Codename (as in the survey): ______________________

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1. Predict the output of both print statements:

```
z = np.arange(6).reshape(3, 2)
print(z.shape)
print(z.sum(axis=0))
```

[3 pts]

Answer:

2. Let X be uniform on $[0, 4]$ (Practice 2, B2). Write down its density $f(x)$, then compute by integration

$$\mathbb{E}[X], \qquad \mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2, \qquad \mathbb{P}(X \in [1, 2]).$$

[3 pts]

Answer:

3. Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad w = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Which of the vectors v, w, u are eigenvectors of A? For each one that is, give its eigenvalue. Justify each answer with a computation (no characteristic polynomial needed — and for a 3×3 matrix you really don't want one). [4 pts]

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.